

Lecture Example 1 - Converting to Standard Form

Tableau Simplex (Big-M) solution report · generated 2026-09-27 13:53

Lecture slides 11-14. Mixed $\geq$ / $\leq$ constraints; used to demonstrate slack, surplus and artificial variables. Solved here to completion as an extension (3 decision variables -> 3-D geometry).

Model

Maximize $Z = 7X_1 - 3X_2 + 5X_3$
subject to
[C1] $X_1 + X_2 + X_3 \geq 9$
[C2] $3X_1 + 2X_2 + X_3 \leq 12$
 $X_1, X_2, X_3 \geq 0$

RESULT

OPTIMAL

$Z^* = 60$

$X_1 = 0$

$X_2 = 0$

$X_3 = 12$

iterations: 3 · rule: dantzig

Findings

- Optimal: row 0 has no negative coefficients. $Z^* = 60$

Contents: standard form · geometry · every tableau with entering/leaving analysis · decision diagram · summary dashboard · independent verification

1. Converting to the standard LP form

1. C1 (Type II, $\geq$): subtract surplus $X4 \geq 0$ (cost 0). Its column is -1 , so it cannot start in the basis $\rightarrow$ an artificial variable is needed.
2. C2 (Type I, $\leq$): add slack $X5 \geq 0$ (unused resource, cost 0) $\rightarrow$ $X5$ is the starting basic variable of this row.
3. C1: add artificial $X6 \geq 0$ with objective coefficient $-M$ (huge penalty) $\rightarrow$ it provides an immediately recognisable starting basic variable.

Maximize $Z = 7X1 - 3X2 + 5X3 - MX6$

subject to

$$X1 + X2 + X3 - X4 + X6 = 9$$

$$3X1 + 2X2 + X3 + X5 = 12$$

$$X1, X2, X3, X4, X5, X6 \geq 0$$

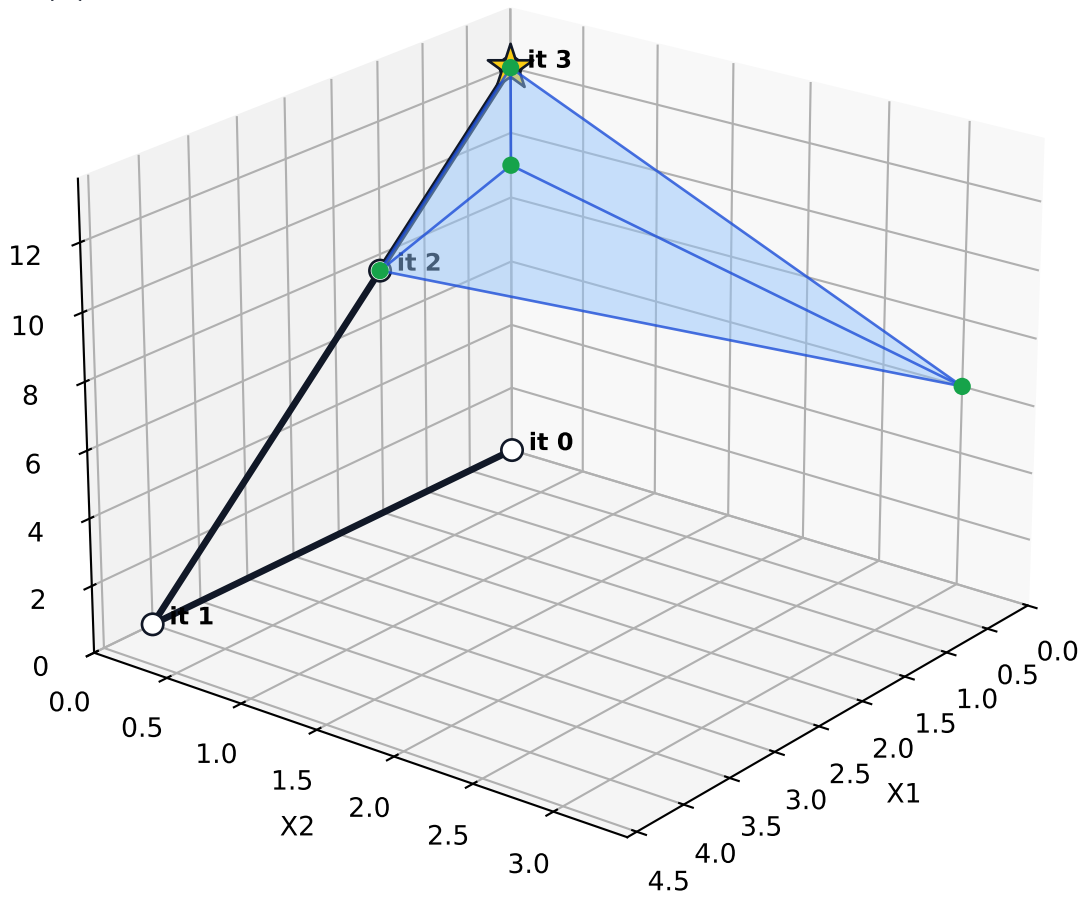
Standard form - coefficient matrix

	X1	X2	X3	X4	X5	X6		RHS
max Z	7	-3	5	0	0	-M		
C1	1	1	1	-1	0	1	=	9
C2	3	2	1	0	1	0	=	12

2. Geometric view: basic solutions, extreme points and the simplex path

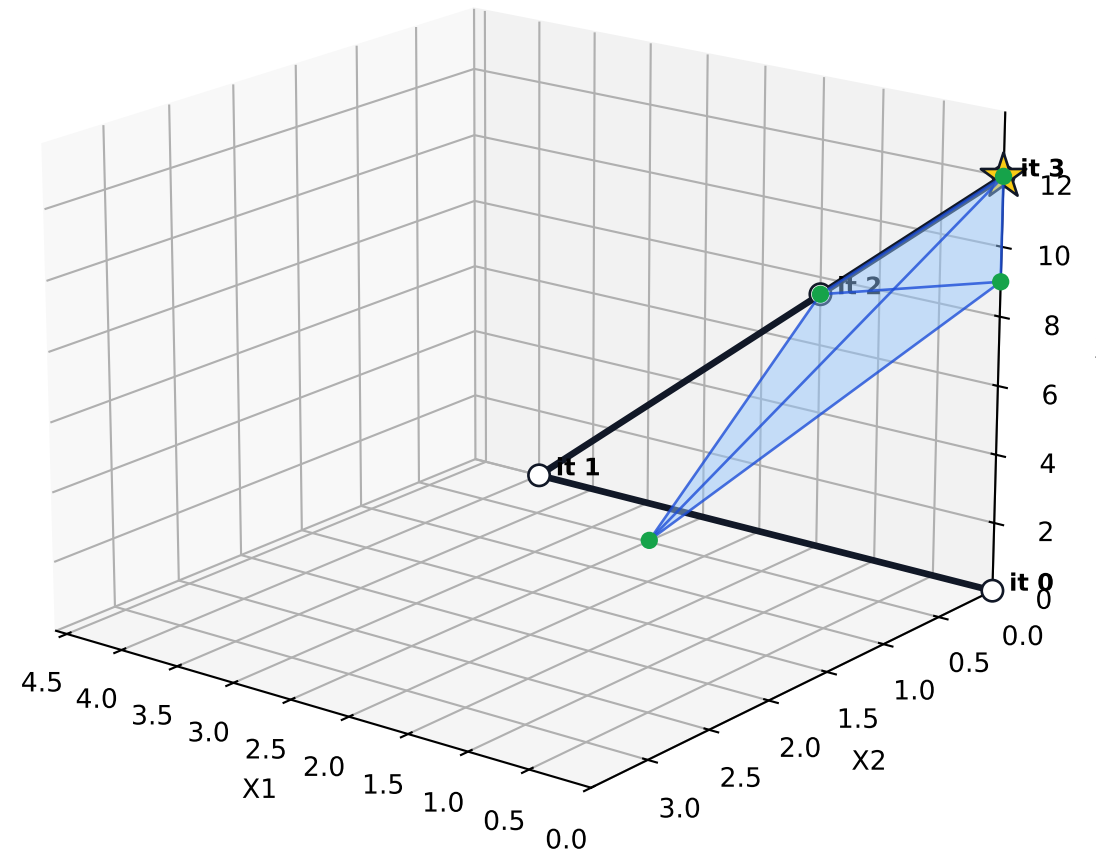
Feasible polytope & simplex path - optimal

- extreme points
- simplex path
- ★ optimum



...same polytope, rotated view

- extreme points
- simplex path
- ★ optimum



3. All basic solutions (m variables basic, the rest 0). Green = feasible = extreme point; red = infeasible; ★ = optimal

Basic variables	X1	X2	X3	Feasible?	Z	X4	X5	Optimal?
X1, X2	-6	15	0	no	-87	0	0	
X1, X3	1.5	0	7.5	yes	48	0	0	
X1, X4	4	0	0	no	28	-5	0	
X1, X5	9	0	0	no	63	0	-15	
X2, X3	0	3	6	yes	21	0	0	
X2, X4	0	6	0	no	-18	-3	0	
X2, X5	0	9	0	no	-27	0	-6	
X3, X4	0	0	12	yes	60	3	0	★
X3, X5	0	0	9	yes	45	0	3	
X4, X5	0	0	0	no	0	-9	12	

Initial tableau

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS
Z	0	1	-7	3	-5	0	0	M	0
X6	1	0	1	1	1	-1	0	1	9
X5	2	0	3	2	1	0	1	0	12

Initial tableau. Basic variables: $X_6 = 9$, $X_5 = 12$. Non-basic (set to 0): X_1 , X_2 , X_3 , X_4 . The values of the basic variables are read directly from the RHS column.

Initial tableau after Big-M preparation

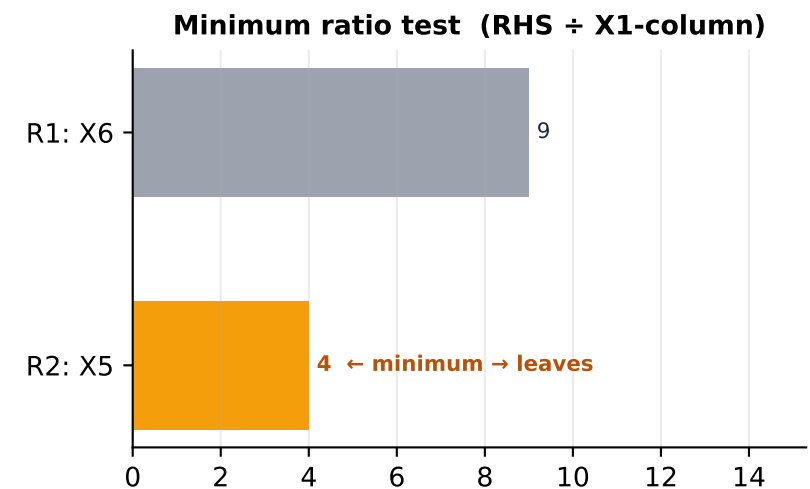
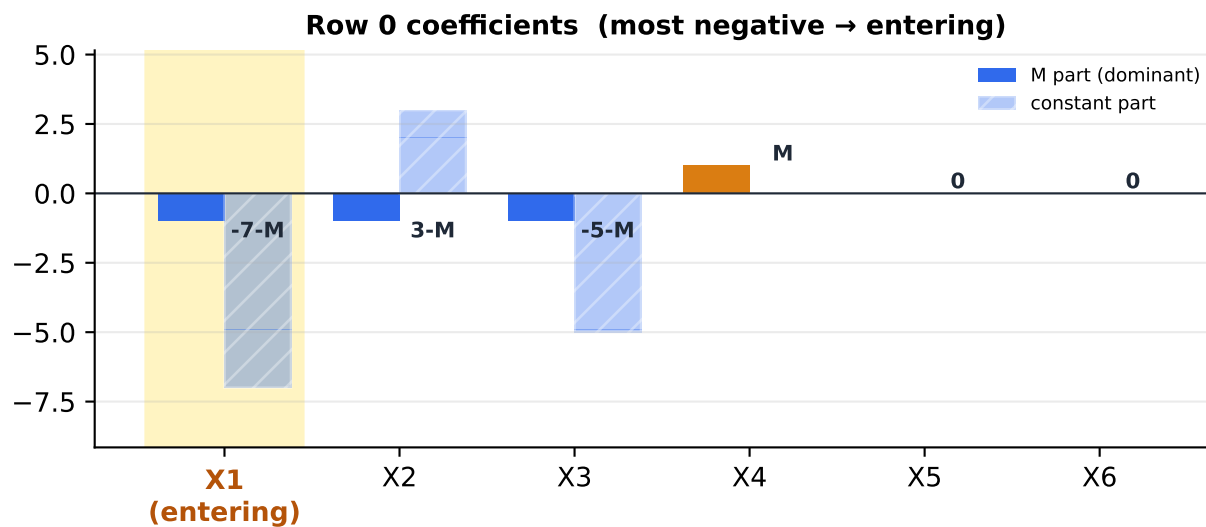
Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS
Z	0	1	-7-M	3-M	-5-M	M	0	0	-9M
X6	1	0	1	1	1	-1	0	1	9
X5	2	0	3	2	1	0	1	0	12

Big-M preparation. the artificial basic variable(s) had coefficient M in row 0. Subtracting M × their rows makes those coefficients zero, so row 0 is ready for the optimality test.

- Subtract M × R1 from R0 (make the coefficient of basic X6 zero)

Iteration 1: entering & leaving variables

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS	Ratio
Z	0	1	-7-M	3-M	-5-M	M	0	0	-9M	
X6	1	0	1	1	1	-1	0	1	9	9
X5	2	0	3	2	1	0	1	0	12	4



Iteration 1 - selecting variables.

- Is the current solution optimal? No - row 0 has at least one negative coefficient.
- Entering basic variable: X1 - it has the largest (in absolute value) negative coefficient in row 0 (-7-M).
- Minimum ratio test (RHS ÷ positive entries of the X1-column): R1: 9, R2: 4.
- Leaving basic variable: X5 (row 2, smallest ratio). Pivot number = 3.

Iteration 1: pivoting (Gauss-Jordan row operations)

(a) pivot row scaled: Divide R2 by the pivot number 3

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS	Ratio
Z	0	1	-7-M	3-M	-5-M	M	0	0	-9M	
X6	1	0	1	1	1	-1	0	1	9	9
X5	2	0	1	0.6667	0.3333	0	0.3333	0	4	4

(b) X1 replaced X5 in the basis. $Z = 28-5M$

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS
Z	0	1	0	7.6667 -0.3333	3.6667 -0.6667	M	2.3333+0.3333	0	28-5M
X6	1	0	0	0.3333	0.6667	-1	-0.3333	1	5
X1	2	0	1	0.6667	0.3333	0	0.3333	0	4

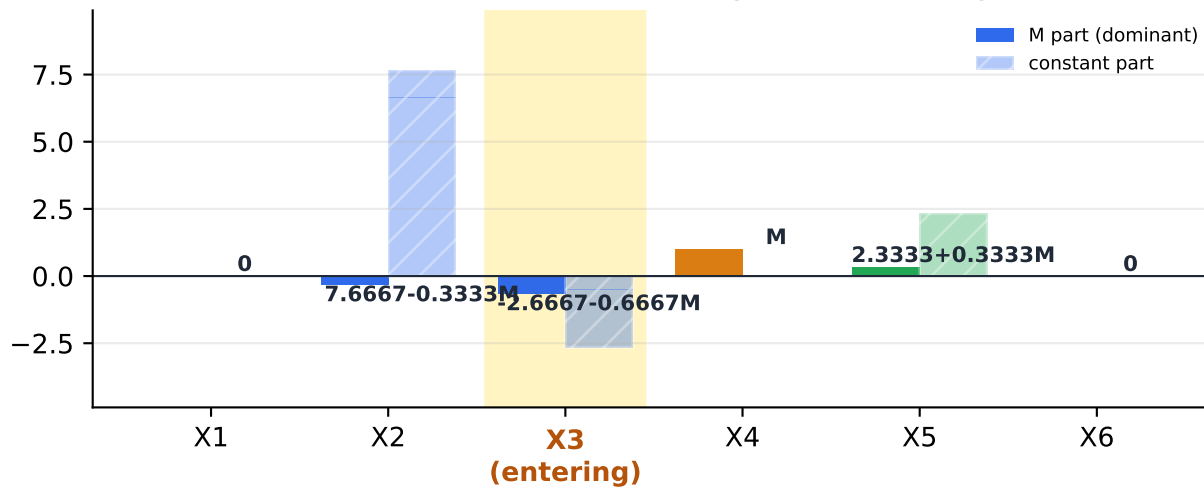
Row operations:

- Add $(7+M) \times R2$ to R0
- Subtract $1 \times R2$ from R1

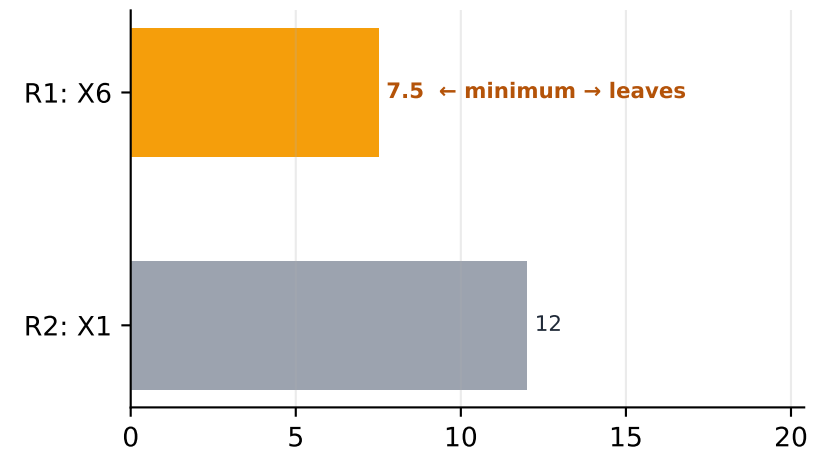
Iteration 2: entering & leaving variables

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS	Ratio
Z	0	1	0	$7.6667-0.3333M$	$-2.6667-0.6667M$	M	$2.3333+0.3333M$	0	$28-5M$	
X6	1	0	0	0.3333	0.6667	-1	-0.3333	1	5	7.5
X1	2	0	1	0.6667	0.3333	0	0.3333	0	4	12

Row 0 coefficients (most negative → entering)



Minimum ratio test (RHS ÷ X3-column)



Iteration 2 - selecting variables.

- Is the current solution optimal? No - row 0 has at least one negative coefficient.
- Entering basic variable: X3 - it has the largest (in absolute value) negative coefficient in row 0 ($-2.6667-0.6667M$).
- Minimum ratio test (RHS ÷ positive entries of the X3-column): R1: 7.5, R2: 12.
- Leaving basic variable: X6 (row 1, smallest ratio). Pivot number = 0.6667.

Iteration 2: pivoting (Gauss-Jordan row operations)

(a) pivot row scaled: Divide R1 by the pivot number 0.6667

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS	Ratio
Z	0	1	0	$7.6667 - 0.3333 \times 0.6667$	$6.6667 - 0.6667 \times 0.6667$	M	$2.3333 + 0.3333 \times 0.6667$	0	$28 - 5M$	
X6	1	0	0	0.5	1	-1.5	-0.5	1.5	7.5	7.5
X1	2	0	1	0.6667	0.3333	0	0.3333	0	4	12

(b) X3 replaced X6 in the basis. Z = 48

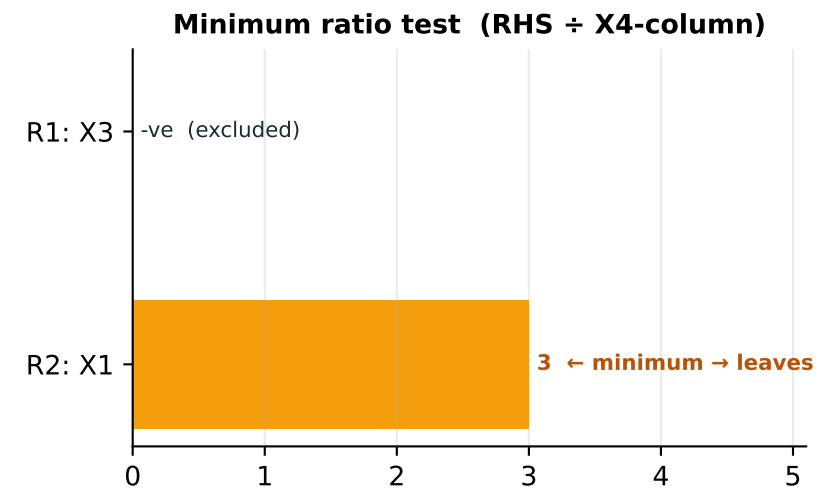
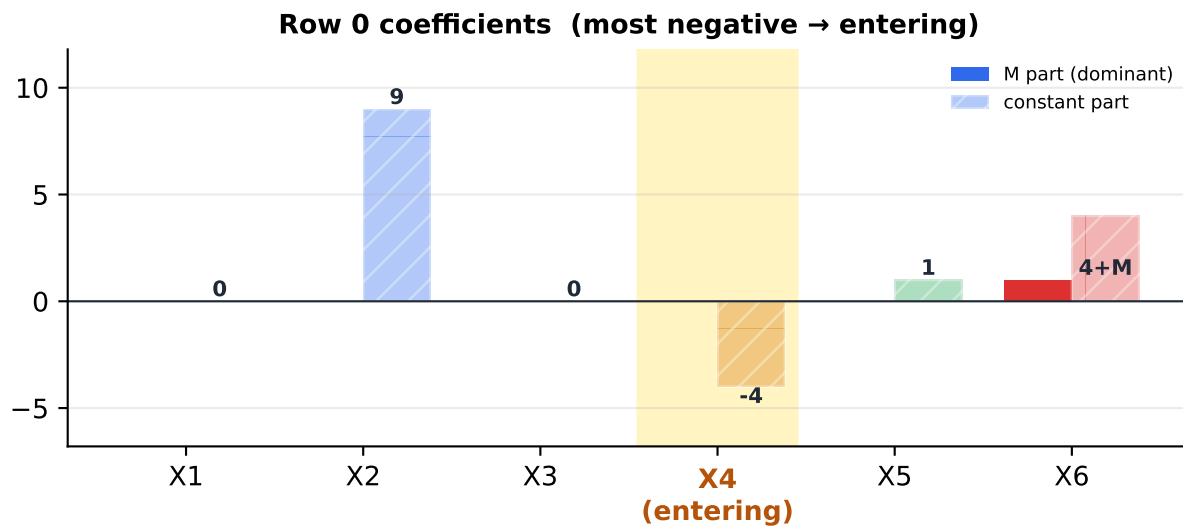
Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS
Z	0	1	0	9	0	-4	1	4+M	48
X3	1	0	0	0.5	1	-1.5	-0.5	1.5	7.5
X1	2	0	1	0.5	0	0.5	0.5	-0.5	1.5

Row operations:

- Add $(2.6667 + 0.6667M) \times R1$ to R0
- Subtract $0.3333 \times R1$ from R2

Iteration 3: entering & leaving variables

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS	Ratio
Z	0	1	0	9	0	-4	1	4+M	48	
X3	1	0	0	0.5	1	-1.5	-0.5	1.5	7.5	-ve
X1	2	0	1	0.5	0	0.5	0.5	-0.5	1.5	3



Iteration 3 - selecting variables.

- Is the current solution optimal? No - row 0 has at least one negative coefficient.
- Entering basic variable: X4 - it has the largest (in absolute value) negative coefficient in row 0 (-4).
- Minimum ratio test (RHS ÷ positive entries of the X4-column): R1: -ve, R2: 3.
- Leaving basic variable: X1 (row 2, smallest ratio). Pivot number = 0.5.

Iteration 3: pivoting (Gauss-Jordan row operations)

(a) pivot row scaled: Divide R2 by the pivot number 0.5

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS	Ratio
Z	0	1	0	9	0	-4	1	4+M	48	
X3	1	0	0	0.5	1	-1.5	-0.5	1.5	7.5	-ve
X1	2	0	2	1	0	1	1	-1	3	3

(b) X4 replaced X1 in the basis. Z = 60

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS
Z	0	1	8	13	0	0	5	M	60
X3	1	0	3	2	1	0	1	0	12
X4	2	0	2	1	0	1	1	-1	3

Row operations:

- Add $4 \times R2$ to R0
- Add $1.5 \times R2$ to R1

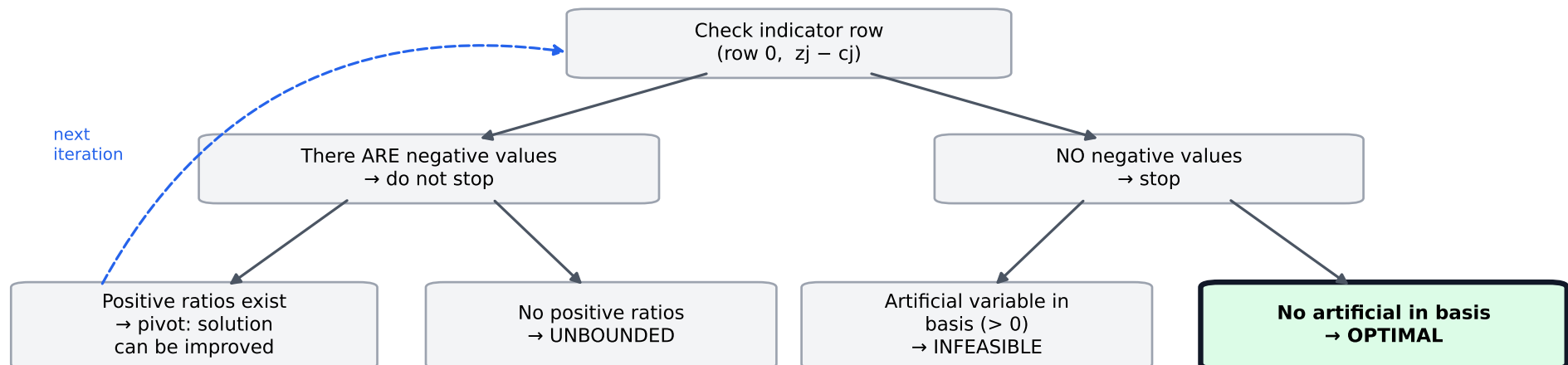
Final tableau - OPTIMAL

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS
Z	0	1	8	13	0	0	5	M	60
X3	1	0	3	2	1	0	1	0	12
X4	2	0	2	1	0	1	1	-1	3

Final check.

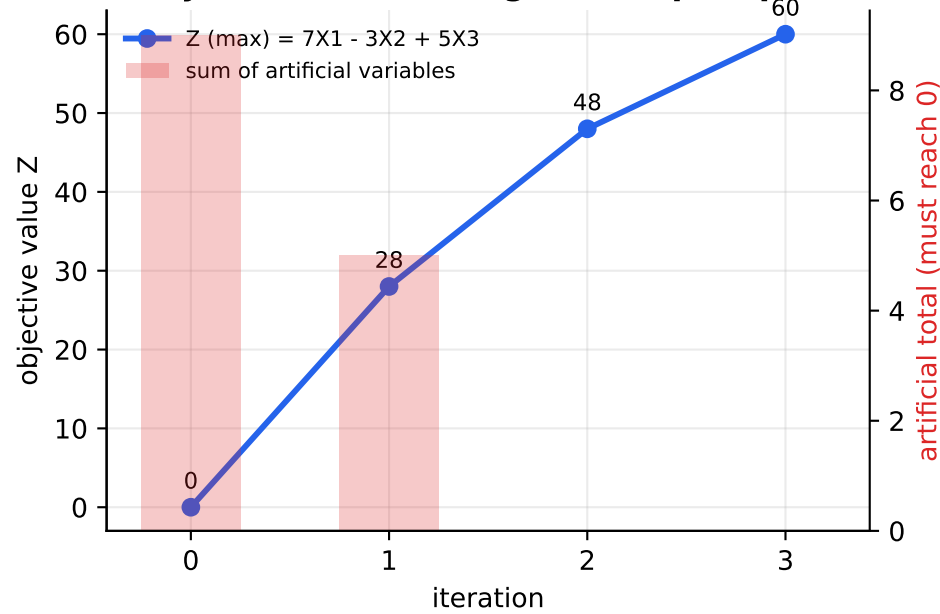
- Optimal: row 0 has no negative coefficients. $Z^* = 60$

Simplex decision diagram - outcome: OPTIMAL

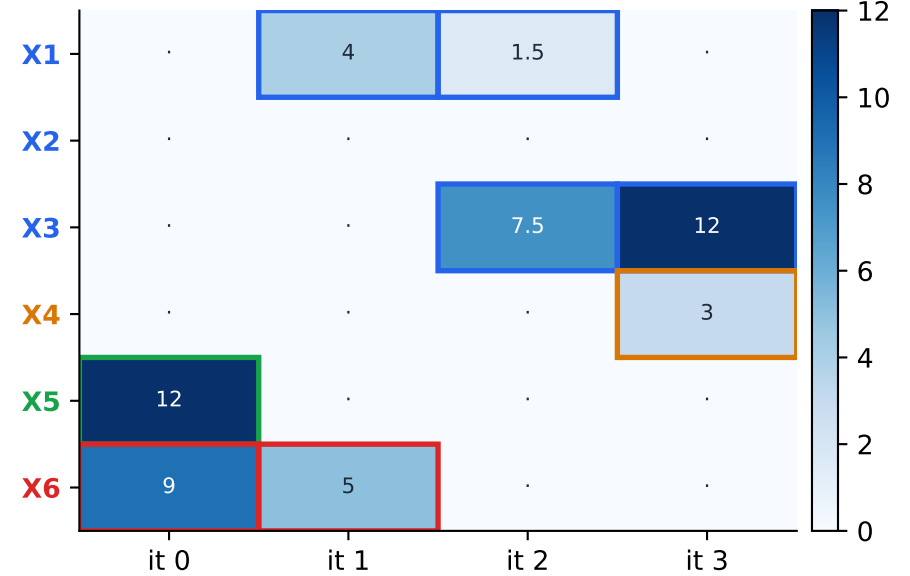


Lecture Example 1 - Converting to Standard Form - OPTIMAL, $Z^* = 60$

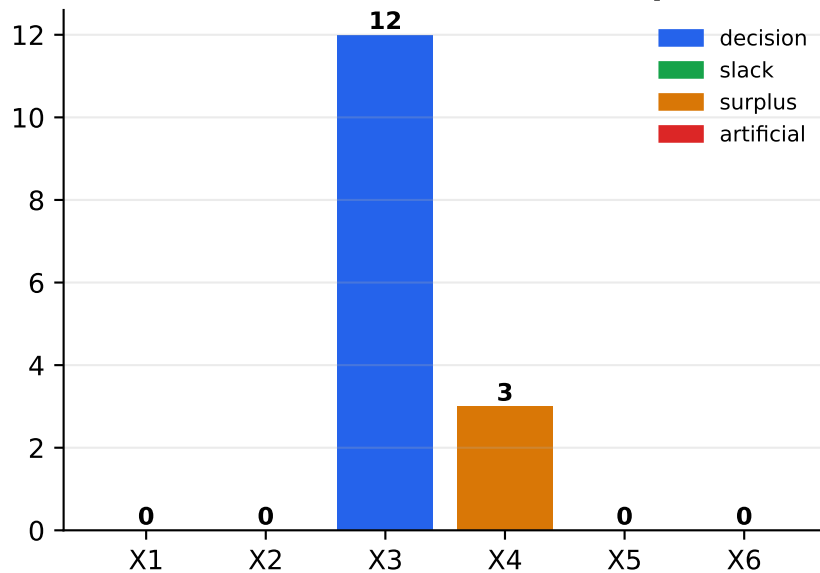
Objective value along the simplex path



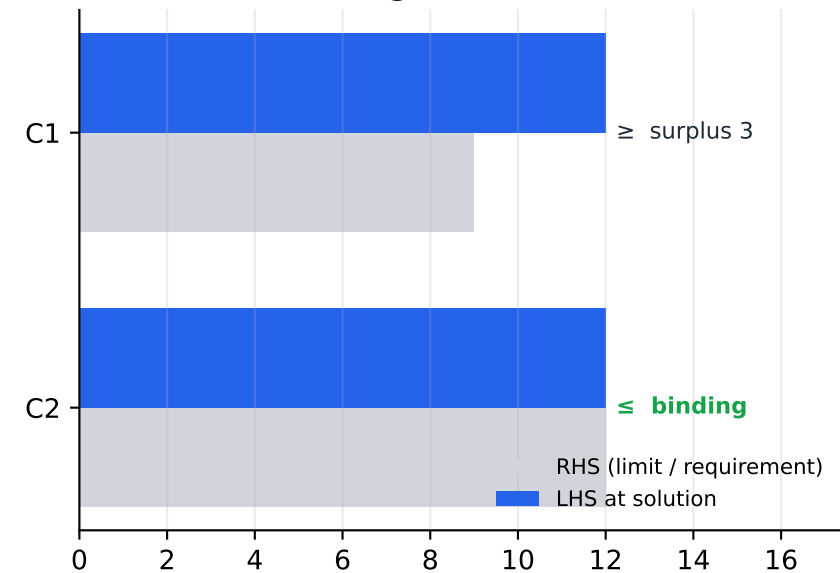
Basis evolution (outlined = basic variable)



Final values of all variables (optimal)



Constraint usage at the final solution



Independent verification with SciPy (HiGHS)

quantity	our tableau simplex	SciPy HiGHS
status	optimal	optimal
Z	60	60
X1	0	0
X2	0	0
X3	12	12

✓ Results agree